

One-page summary — the five most consequential decisions

- 1. Results-day reads are served from a pre-materialized static tier, not the live database (ADR-001).** Every candidate payload and school export is generated and published to object storage/CDN before the announced release time, so the ~150x results-day traffic spike is absorbed at flat cost instead of by scaling the database. It costs the Head of Operations the "instant everywhere" amendment experience they asked for: a late correction is not visible until the next scheduled refresh.
- 2. Results are append-only, versioned records — never updated in place (ADR-002).** Every amendment creates a new immutable version linked to the officer and the evidence, satisfying the Regulator's 50-year point-in-time reproducibility requirement. It costs the Head of Operations the literal ability to retroactively rewrite a certificate an employer has already verified — the request that, taken literally, was impossible.
- 3. PostgreSQL, partitioned by examination sitting, not sharded (ADR-003, ADR-004).** A single relational primary with read replicas fits the actual write volume in this brief, and partitioning by sitting keeps every required query flat-cost as the archive grows to 216,000,000 rows. It costs a future need to query across many sittings for one candidate at once, which no stakeholder in this brief has asked for.
- 4. Each of the three external consumers gets its own protocol and authentication (ADR-005, ADR-006).** Candidates on 2G get USSD/SMS with no TCP/TLS handshake cost; the employers' association gets mutual TLS as a contracted machine client; schools get bearer tokens sized for managing 21,000 accounts. It costs three integration surfaces to build and operate instead of one.
- 5. The PIN double-charge is fixed entirely on Takarda's side via a reserve/confirm state machine (ADR-007).** A client-supplied idempotency key lets Takarda dedupe retries before ever calling the payment partner, and a reconciliation job resolves any charge whose outcome was unknown at the time. It costs Takarda ongoing ownership of a state machine and nightly reconciliation job that, ideally, the payment partner would have handled itself.